Lecture 4

Thursday, 19 January 2023 10:30 AM

Simple KE functional: $\frac{E}{N} = e_{kin}(\rho) = \frac{\pi^2\hbar^2}{6m}\rho^2$

(for non-interacting fermions in a box)

↓

energy per per particle

→ $\rho(x)\cdot dx$ is no. of particles in length dx

→ $e_{kin}(\rho)(\rho\cdot dx)$ is KE in length dx

→ Total energy is $E_{KE}[\rho(x)] = \int e_{kin}(\rho)\cdot\rho\,dx$ over all space } general KE expression

If there is potential energy or interactions, $E_{kin}[\rho(x)] = \int e_{tot}(\rho)\,\rho\cdot dx$

Ex: Reproduce PIB energies for N particles

$$E[\rho] = \int_0^L \frac{\pi^2\hbar^2}{6m}\rho^3\cdot dx \quad ; \text{ Constraint: } \int_0^L \rho(x)\cdot dx = N$$

(to avoid trivial solution)

Use Lagrange multipliers:

define $F[\rho] \equiv E[\rho] - \mu\left(\int_0^L \rho(x)\cdot dx - N\right)$

↑ Lagrange multiplier

Note: $\frac{\partial}{\partial\rho}\int F(\rho,\nabla\rho)\cdot dr = \frac{\partial f}{\partial\rho} - \frac{\partial f}{\partial\nabla\rho}$

function of ρ & grad ρ

Then minimize

$$\frac{\partial F(\rho)}{\partial\rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 - \mu\left(\frac{\partial}{\partial\rho}\int\rho\,dx - \frac{\partial N}{\partial\rho}\right)$$

($\int\rho\,dx \to 1$, $N \to 0$)

$$\therefore \mu = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2$$

↓

$$\int_0^L \sqrt{\frac{2m\mu}{\pi^2\hbar^2}}\cdot dx = N \Rightarrow \sqrt{\frac{2m\mu}{\pi^2\hbar^2}} = \frac{N}{L}$$

$$\boxed{\rho_{min} = \frac{N}{L}}$$

$$\therefore \boxed{E_{min} = \frac{\pi^2 \hbar^2}{6m} \frac{N^3}{L^3} L}$$

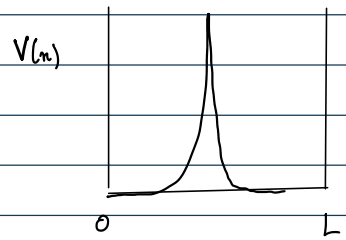
This approximation improves for large N:

N	E_{exact}	E_{DFT}	$\frac{DFT}{exact}$
1	1	$1/3$	0.33
4	30	$64/3$	8.71
20	2870	$8000/3$	0.93

Principle: DFT gives more accurate results for systems that are similar to the systems for which the functional was derived

However, results are surprisingly good for different system types, as well

Example 2



δ function spike

Properties of δ: 1) $\int_0^\infty \delta(x)\,dx = 1$ (thin & tall)

2) $\int_0^L f(x)\,\delta\left(x-\frac{L}{2}\right)dx = f\left(\frac{L}{2}\right)\int_0^L \delta\left(x-\frac{L}{2}\right)dx = f\left(\frac{L}{2}\right)$

New functional: $E[\rho] = \int_0^L \frac{\pi^2\hbar^2}{6m}\rho^3\,dx + \underbrace{\alpha\int \delta\left(x-\frac{L}{2}\right)\rho\,dx}_{\text{external PE}}$

Lagrange functional:

$$F[\rho] \equiv E[\rho] - \mu\left(\int_0^L \rho\,dx - N\right)$$

$$\frac{\partial F}{\partial \rho} = -\mu + \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + \frac{\partial}{\partial \rho}\left(\rho\left(\tfrac{L}{2}\right)\cdot\alpha\right) \to 0$$

$$\therefore \rho_{min} = \sqrt{\frac{2m\mu}{\pi^2\hbar^2}} = \frac{N}{L} \text{ constant}$$

S does not affect density

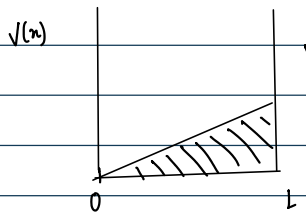
$$E_{min} = \frac{\pi^2\hbar^2}{6m}\left(\frac{N}{L}\right)^3 L + \frac{\alpha N}{L}\int_0^L \delta(x-\tfrac{L}{2})\cdot dx \quad (\int \delta = 1)$$

$$\therefore E_{min} = \frac{\pi^2\hbar^2}{6m}\left(\frac{N}{L}\right)^3 L + \frac{\alpha N}{L}$$

S does add $\frac{\alpha}{L}$ per particle → (α = constant)

physically an obstacle (or nucleus) in the center will act as a S function (it will affect the energy everytime an e^- crosses it in the particle in box example)

Example 3



$$V_{ext}(x) = \frac{V_{max}}{L}x$$

$$E_{tot}[\rho] = \frac{\pi^2\hbar^2}{6m}\rho^2 + \frac{x}{L}V_{max}$$

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_0^L \rho^3\cdot dx + \int_0^L \frac{V_{max}}{L}x\cdot\rho\,dx$$

Lagrange multiplier:

$$\frac{\partial F}{\partial \rho} = 0 = \frac{\pi^2\hbar^2}{2m}\rho_{min}^2 + \frac{V_{max}}{L}x - \mu$$

$$\therefore \rho_{min} = \sqrt{\frac{2m}{\pi^2\hbar^2}\left(\mu - \frac{V_{max}}{L}x\right)}$$ (ρ_{min} is not constant)

Finding μ: $\int_0^L \rho_{min}\cdot dx = N$

$$\int_0^L \sqrt{\frac{2m}{\pi^2\hbar^2}\left(\mu - \frac{V_{max}}{L}x\right)}\cdot dx = N$$

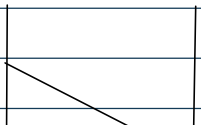
u-sub: $u = \mu - \frac{V_{max}}{L}x$

$$(\mu - V_{max})^{3/2} - \mu^{3/2} = -\frac{3}{2}\frac{V_{max}N}{L}\sqrt{\frac{\pi^2\hbar^2}{2m}}$$

Either solve numerically or solve for small $\frac{V_{max}}{\mu}$.

$$\mu \approx \frac{\pi^2\hbar^2}{2m}\frac{N^2}{L^2} + O(V_{max})$$

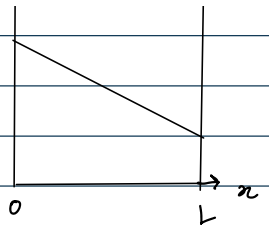
$$\rho_{min}[x] = \frac{N}{L}\left[1 - \frac{V_{max}x}{L}\frac{2mL^2}{\pi^2\hbar^2N^2}\right]$$



Electric field pushes on particle, less probability for it to be on

$$\rho_{min}[x] = \frac{N}{L}\left[1 - \frac{V_{max}\,x}{L}\,\frac{2mL^2}{\pi^2\hbar^2N^2}\right]$$

$\gamma = \frac{N}{L}$ $\quad \beta = \frac{V_{max}}{L}\frac{2mL^2}{\pi^2\hbar^2N^2}$



Electric field pushes on particle, less probability for it to be on that side of the box

$$\rho = \gamma(1-\beta x)$$

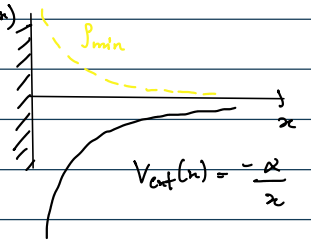
$$E[\rho_{min}] = \frac{\pi^2\hbar^2}{6m}\int_0^L \gamma^3(1-\beta x)^3\,dx + \frac{\gamma V_{max}}{L}\int_0^L x(1-\beta x)\,dx$$

$$= -\frac{1}{\beta}\frac{\pi^2\hbar^2}{6m}\frac{\gamma^3}{4}\underbrace{\left[(1-\beta L)^4 - 1\right]}_{\approx -4\beta L \text{ (for small } \beta)} + \frac{V_{max}}{L}\gamma\left(\frac{L^2}{2} - \beta\frac{L^3}{3}\right)$$

$$\boxed{E_{min} = \frac{\pi^2\hbar^2}{6mL^2}N^3 + \frac{1}{2}V_{max}N} \quad \text{for small } V_{max}$$

↳ So we add $\frac{1}{2}V_{max}$ to energy of each particle (average)

HW preview: V(x)



Coulomb potential

$\mu < 0$ → need absolute values. (replace $\mu = -|\mu|$)

Answer: $E[\rho] = \frac{\pi^2\hbar^2}{6m}\int_0^\infty \rho^3\,dx - \alpha\int_0^\infty \frac{\rho(x)}{x}\,dx$

$$0 = \frac{\partial F}{\partial \rho} \rightarrow \quad \rho_{min}(x) = \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2}\left(\mu + \frac{\alpha}{x}\right)^{1/2}$$

$$\Rightarrow N = \int_0^\infty \rho_{min}\,dx = \left(\frac{2m}{\pi^2\hbar^2}\right)^{1/2}\int_0^{\alpha/|\mu|}\left(\frac{\alpha/|\mu|}{x} - 1\right)^{1/2}dx\;|\mu|^{1/2}$$

ρ can never be imaginary!

So truncate domain to keep ρ_{min} real $\left(\frac{\alpha}{|\mu|}\right)$

$$\mu = -\frac{m}{2\hbar^2}\frac{\alpha^2}{N^2}$$

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}N^3 + \frac{\pi^2\hbar^2}{6m}\left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2}|\mu|^{\frac{3}{2}}\int_0^{\alpha/|\mu|}\left(\frac{\alpha/|\mu|}{x} - 1\right)^{3/2}dx$$

$$E_{min}[\rho_{min}] = \frac{\pi^2\hbar^2}{6mL^2}N^3 + \frac{\pi^2\hbar^2}{6m}\int_0^{\alpha/|\mu|}\left(\frac{2m}{\pi^2\hbar^2}\right)^{3/2}|\mu|^{\frac{3}{2}}\left(\frac{\alpha/|\mu|}{x}-1\right)^{3/2}\cdot dx$$

diverges (since we are modelling an atom in 1D)

↓

Larger N means that particles will start to pack in the lower potential regime → nowhere for them to go

↓

unphysically packed

Thomas-Fermi Atom

Interactions b/w particles

1) Coulomb : classic electrostatic interactions

$E \sim \frac{e^2}{4\pi\varepsilon_0 x}$ repulsion b/w electrons

Electrostatic functional:

$$E_{coul} = \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\iint\frac{\rho(x)\,\rho(x')}{(x-x')}\,dx\,dx'$$

dummy var

avoid double counting

2) Exchange : purely QM property/interaction (attractive and b/w charged particles)

↳ Fully accounted for in TISE, but DFT requires extra terms

Simple approximation w/ dimensional analysis

$$E_{exch} \sim \int\frac{e^2}{\varepsilon_0}\rho^n\cdot dx$$

$n=2$ gives dimensional consistency

Putting it together: (in 1D)

$$E[\rho] = \frac{\pi^2\hbar^2}{6m}\int \rho^3\,dx + \frac{1}{2}\frac{e^2}{4\pi\varepsilon_0}\iint \frac{\rho(x)\rho(x')}{|x-x'|}\,dx\,dx' - C_x\frac{e^2}{4\pi\varepsilon_0}\int \rho^2\,dx + \int V_{ext}\,\rho\,dx$$

V_{ext} could be $-\frac{\alpha}{x}$, $\frac{1}{2}kx^2$, ...